53.5: Density-dependent factors regulate population growth

What environmental factors keep populations from growing indefinitely? Why are some populations fairly stable in size, while others are not?

Answers to such questions can be important in practical applications. Farmers may want to reduce the abundance of insect pests or stop the growth of a weed that is spreading rapidly. Conservation ecologists need to know what environmental factors create favorable feeding or breeding habitats for endangered species, such as the white rhinoceros and the whooping crane. Overall, whether seeking to reduce the size of an unwanted population or increase the size of one that is endangered, it is helpful to understand factors that affect population abundance.

Population Change and Population Density

To understand why a population stops growing when it reaches a certain size, ecologists study how the rates of birth, death, immigration, and emigration change as population density rises. If immigration and emigration offset each other, then a population grows when the birth rate exceeds the death rate and declines when the death rate exceeds the birth rate.

A birth rate or death rate that does *not* change with population density is said to be **density independent**. For example, researchers found that the mortality of dune fescue grass (*Vulpia fasciculata*) is mainly due to physical factors that kill similar proportions of a local population, regardless of its density. Drought stress that arises when the roots of the grass are uncovered by shifting sands is a density-independent factor that can kill these plants. In contrast, a death rate that increases with population density or a birth rate that falls with rising density is said to be **density dependent**. Researchers found that reproduction by dune fescue declines as population density increases, in part because water or nutrients become more scarce. Thus, the key factors affecting birth rate in this population are density dependent, while death rate is largely determined by density-independent factors. For situations in which immigration and emigration offset each other, **Figure 53.15** shows how the combination of density-dependent reproduction and density-independent mortality can stop population growth (as in species such as dune fescue).

Figure 53.15

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Determining equilibrium for population density.
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A graph plots birth or death rate per capita versus population density.

In this example, the birth rate changes with population density, while the death rate is constant; immigration and emigration rates are assumed to be equal. At the equilibrium density (Q), the birth and death rates are equal. As a result, the number of individuals added to the population equals the number removed from it, and the population stops changing in size.

Graph of birth or death rate per capita versus population density. The x-axis is increasing population density and the y-axis is increasing birth or death rate per capita. Equilibrium density (Q) is the Intersection of the density-independent death rate (m) line and the density-dependent birth rate b. line. The density-independent death rate (m) is a flat horizontal line. The density dependent birth rate b. is a line with a negative slope running through Q and starting high on the birthrate per capita axis and sloping towards the low values on the y-axis as population density increases. When population density is low, b is greater than m. As a result, the population grows until the density reaches Q. When population density is high, m is greater than b, and the population shrinks until the density reaches Q.

DRAW IT *Redraw this figure for the case where the birth and death rates are both density dependent, as occurs for many species.*

*For suggested answer, see*Appendix A.**

Variation in density-independent factors such as temperature and precipitation can cause dramatic changes in population size. For example, a drought or heat wave can cause a sharp increase in mortality rates, causing the abundance of a population to plummet. Note, however, that a density-independent factor cannot consistently cause a population to decrease in size when it is large or increase in size when it is small—only a density-dependent factor can consistently cause such changes. With this in mind, a population is said to be *regulated* when one or more density-dependent factors cause its size to decrease when large (or increase when small).

BioFlix® Animation

Density Dependence

image

Play

image

Mechanisms of Density-Dependent Population Regulation

Without some type of negative feedback between population density and the rates of birth and death, a population would never stop growing. But no population can increase in size indefinitely. Ultimately, at large population sizes, negative feedback is provided by density-dependent regulation, which halts population growth through mechanisms that reduce birth rates or increase death rates. For example, a study of kelp perch (*Brachyistius frenatus*) populations showed that the fish's death rate rose proportionally as its density increased **(Figure 53.16)**. This occurred because at high densities, the fish ran out of safe spaces in the kelp where they could hide from predators. Several other mechanisms that can cause density-dependent regulation are described in **Figure 53.17**.

Figure 53.16

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An example of density-dependent regulation.
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A graph plots proportional mortality of kelp perch versus kelp perch density. Proportional mortality of kelp perch increases with kelp perch density.

At high population densities, the death rate of kelp perch increased because they ran out of spaces to hide from predators.

Graph of proportional mortality of kelp perch versus kelp perch density. The x-axis is kelp perch density (number per plot), ranging from 0 to 60 with increments every 10. The y-axis is proportional mortality, ranging from 0 to 1.0 with increments every 0.2. Proportional mortality of kelp perch increases with kelp perch density. The graph rises from (10, 0.2) to (20, 0.35) to (40, 0.45) to (60, 0.7). All values are estimated.

Figure 53.17

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Exploring Mechanisms of Density-Dependent Regulation
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Exploring mechanisms of density-dependent regulation.

As population density increases, many density-dependent mechanisms slow or stop population growth by decreasing birth rates or increasing death rates. Mechanisms include competition, disease, predation, territoriality, intrinsic factors, and toxic wastes.

Competition for resources. Increasing population density intensifies competition for nutrients and other resources, reducing reproductive rates. Farmers minimize the effect of resource competition on the growth of wheat (Triticum aestivum) and other crops by applying fertilizers to reduce nutrient limitations on crop yield. Image: wheat plants crowded together in a field.

Disease: If the transmission rate of a disease increases as a population becomes more crowded, then the disease's impact is density dependent. In humans, the respiratory diseases influenza (flu) and tuberculosis are spread through the air when an infected person sneezes or coughs. Both diseases strike a greater percentage of people in densely populated cities than in rural areas. Image: a crowd of people walking down steps; one person is wearing a respirator mask.

Predation. Predation can be an important cause of density-dependent mortality if a predator captures more food as the population density of the prey increases. As a prey population builds up, predators may also feed preferentially on that species. Population increases in the collared lemming (Dicrostonyx groenlandicus) lead to density-dependent predation by several predators, including the snowy owl (Bubo scandiacus). Image: a snowy owl.

Territoriality. Territoriality can limit population density when space becomes the resource for which individuals compete. Cheetahs (Acinonyx jubatus) use a chemical marker in urine to warn other cheetahs of their territorial boundaries. The presence of surplus, or nonbreeding, individuals is a good indication that territoriality is restricting population growth. Image: a solitary cheetah.

Intrinsic factors. Intrinsic physiological factors can regulate population size. Reproductive rates of white-footed mice (Peromyscus leucopus) in a field enclosure can drop even when food and shelter are abundant. This drop in reproduction at high population density is associated with aggressive interactions and hormonal changes that delay sexual maturation and depress the immune system. Image: white-footed mice with young.

Toxic wastes. Yeasts, such as the brewer's yeast Saccharomyces cerevisiae, are used to convert carbohydrates to ethanol in winemaking. The ethanol that accumulates in the wine is toxic to yeasts and contributes to density-dependent regulation of yeast population size. The alcohol content of wine is usually less than 13% because that is the maximum concentration of ethanol that most wine-producing yeast cells can tolerate. Image: brewer's yeast, scale 5 micrometers.

These various examples of population regulation by negative feedback show how increased densities cause population growth rates to decline by affecting reproduction, growth, and survival. Although negative feedback helps explain why populations stop growing, it does not address why some populations fluctuate dramatically while others remain relatively stable.

Population Dynamics

Like the piping plover (see Figure 53.1), all populations show some fluctuation in size. Such population fluctuations from year to year or place to place, called **population dynamics**, are influenced by many factors and in turn affect other species. For example, fluctuations in fish populations affect populations of

seabirds that eat fish. The study of population dynamics focuses on the complex interactions between biotic and abiotic factors that cause variation in population sizes.

Stability and Fluctuation

Populations of large mammals were once thought to remain relatively stable, but long-term studies have challenged that idea. For instance, the moose population on Isle Royale in Lake Superior has fluctuated substantially since around 1900. At that time, moose from the Ontario mainland (25km away) colonized the island, perhaps by walking across the lake when it was frozen. Wolves, which rely on moose for most of their food, reached the island around 1950 by walking across the frozen lake. The lake has not frozen over since the early 1950s, and both populations appear to have been isolated from immigration and emigration since then. Despite this isolation, the moose population experienced two major increases and collapses during the last 50 years **(Figure 53.18)**.

Figure 53.18

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    Fluctuations in moose and wolf population sizes on Isle
Royale, 1959–2011.
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Graph of number of wolves and number of moose versus year. Both populations show fluctuations, with dramatic peaks and declines.

See table next tab

BioFlix® Animation

Predator-Prey Interactions

image

Play

image

What factors cause the size of the moose population to change so dramatically? Harsh weather, particularly cold winters with heavy snowfall, can weaken moose and reduce food availability, decreasing the population size. When moose numbers are low and the weather is mild, food is readily available and the population grows quickly. Conversely, when moose numbers are high, factors such as predation and an increase in the density of ticks and other parasites cause the population to shrink. The effects of some of these factors can be seen in Figure 53.18. The first major collapse coincided with a peak in the numbers of wolves from 1975 to

1980. The second major collapse, around 1995, coincided with harsh winter weather, which increased the energy needs of the moose and made it harder for them to find food under the deep snow.

Population Cycles: *Scientific Inquiry*

While many populations fluctuate at unpredictable intervals, others undergo regular boom-and-bust cycles. Some small herbivorous mammals, such as voles and lemmings, tend to have 3- to 4-year cycles, while some birds, such as ruffed grouse and ptarmigans, have 9- to 11-year cycles.

One striking example of population cycles is the roughly 10-year cycling of snowshoe hares (*Lepus americanus*) and lynx (*Lynx canadensis*) in the far northern forests of Canada and Alaska. Lynx are predators that feed predominantly on snowshoe hares, so lynx numbers might be expected to rise and fall with the numbers of hares **(Figure 53.19)**. But why do hare numbers rise and fall in approximately 10-year cycles? Two main hypotheses have been proposed. First, the cycles may be caused by food shortage during winter. Hares eat the terminal twigs of small shrubs such as willow and birch in winter, although why this food supply might cycle in 10-year intervals is uncertain. Second, the cycles may be due to predator-prey interactions. Many predators other than lynx eat hares, and they may overexploit their prey.

Figure 53.19

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Population cycles in the snowshoe hare and lynx.
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A graph plots number of hares and number of lynx versus time. Both populations show cyclical fluctuations, with lynx population peaks following snowshoe hare population peaks.

Population counts are based on the number of pelts sold by trappers to the Hudson Bay Company.

Data are as follows:

Year	Number of hares (thousands)	Number of lynxes (thousands)
1850	40	2
1860	85	3
1870	160	4
1872	80	6
1873	10	2

1874	20	1
1880	90	3
1881	80	4
1885	15	1.5
1890	85	4
1900	10	1
1905	70	4
1906	65	5
1910	30	2.5
1911	35	1
1915	80	3
1916	45	4
1920	80	3
1925	40	5
1930	15	2.5
1931	20	1

INTERPRET THE DATA *What do you observe about the relative timing of the peaks in lynx numbers and hare numbers? What might explain this observation?*

Let's consider the evidence for the two hypotheses. If hare cycles are due to winter food shortage, then the cycles should stop if extra food is provided to a field population. Researchers conducted such experiments in the Yukon for 20-year—over two hare cycles. They found that hare populations in the areas with extra food increased about threefold in density but continued to cycle in the same way as the unfed control populations. Therefore, food supplies alone do not cause the hare cycles shown in Figure 53.19, so we can reject the first hypothesis.

To study the effects of predation, ecologists used radio collars to track individual hares to determine why they died. Predators, including lynx, coyotes, hawks, and owls, killed 95% of the hares in such studies. None of the hares appeared to have died of starvation. These data support the second hypothesis. When ecologists

set up electric fences to exclude predators from certain areas, the collapse in survival that normally occurs in the decline phase of the cycle was nearly eliminated. Overexploitation by predators thus seems to be an essential part of snowshoe hare cycles; without predators, it is unlikely that hare populations would cycle in northern Canada.

Immigration, Emigration, and Metapopulations

So far, our discussion of population dynamics has focused mainly on the contributions of births and deaths. However, immigration and emigration also influence populations. When a population becomes crowded and resource competition increases, emigration often increases.

Immigration and emigration are particularly important when a number of local populations are linked, forming a **metapopulation**. Local populations in a metapopulation can be thought of as occupying discrete patches of suitable habitat in a sea of otherwise unsuitable habitat. Such patches vary in size, quality, and isolation from other patches, factors that influence how many individuals move among the populations. If one population becomes extinct, the patch it occupied may be recolonized by immigrants from another population.

The Glanville fritillary (*Melitaea cinxia*) illustrates the movement of individuals between populations. On average, this butterfly is found in about 500 meadows across the Åland Islands of Finland, but its potential habitat in the islands is much larger, approximately 4,000 suitable patches. New populations of the butterfly regularly appear and existing populations become extinct, constantly shifting the locations of the 500 colonized patches **(Figure 53.20)**. The species persists in a balance of local extinctions and recolonizations.

Figure 53.20

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The Glanville fritillary: a metapopulation.
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A map of local populations of the Glanville fritillary on the Aland Islands of Finland. Suitable habitat is scattered throughout the islands, with more unoccupied habitat patches than occupied patches.

On the Åland Islands, local populations of this butterfly (filled circles) are found in only a fraction of the suitable habitat patches at any given time. Individuals can move between local populations and colonize unoccupied patches (open circles).

An individual's ability to move between populations depends on a number of factors, including its genetic makeup. One gene that has a strong effect on the Glanville fritillary's ability to move is *Pgi*, which codes for the enzyme phosphoglucoisomerase. This enzyme catalyzes the second step of glycolysis (see Figure 9.8),

and its activity correlates with the rate of CO_2 production from respiration by the butterflies. Ecologists studied butterflies known to be heterozygous or homozygous for a single nucleotide polymorphism in *Pgi*. They tracked the movements of individual butterflies using radar and transponders attached to the butterflies that emit an identifying signal. Butterfly movements ranged widely, from 10 m to 4 km, in 2-hour periods. Heterozygous individuals flew more than twice as far in the morning and at lower ambient temperatures than did homozygous individuals. The results indicated a fitness advantage to the heterozygous genotype in low temperatures and a greater likelihood of heterozygotes colonizing new locations in the metapopulation.

A Glanville fritillary (Melitaea cinxia) wearing a tracking transponder.

A Glanville fritillary (Melitaea cinxia) wearing a tracking transponder.

The metapopulation concept underscores the significance of immigration and emigration for the Glanville fritillary and many other species. It also helps ecologists understand population dynamics and gene flow in patchy habitats, providing a framework for the conservation of species living in a network of habitat fragments and reserves.